

Dataset Description Report – Heart Disease Dataset

1. Dataset Overview

The Heart Disease dataset is a medical dataset commonly used for machine learning tasks related to cardiovascular disease prediction. It contains clinical and demographic information about patients and is suitable for classification, regression, and clustering tasks.

- **Dataset Name:** Heart Disease Dataset
 - **Source:** Kaggle
 - **File Format:** CSV
 - **Number of Records (Rows):** 303
 - **Number of Features (Columns):** 14
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2. Dataset Purpose

The dataset is primarily designed for **binary classification**, where the goal is to predict whether a patient has heart disease or not. However, it can also be adapted for: - **Regression tasks** (e.g., predicting patient age) - **Clustering tasks** to discover patient subgroups based on clinical features

3. Features Description

Feature	Description
age	Age of the patient in years
sex	Gender (1 = male, 0 = female)
cp	Chest pain type
trestbps	Resting blood pressure (mm Hg)
chol	Serum cholesterol (mg/dl)
fbs	Fasting blood sugar (>120 mg/dl)
restecg	Resting electrocardiographic results
thalach	Maximum heart rate achieved
exang	Exercise-induced angina (1 = yes, 0 = no)
oldpeak	ST depression induced by exercise

Feature	Description
slope	Slope of the peak exercise ST segment
ca	Number of major vessels colored by fluoroscopy
thal	Thalassemia type
target	Heart disease status (1 = presence, 0 = absence)

4. Data Types

- **Numerical Features:** age, trestbps, chol, thalach, oldpeak
- **Binary Features:** sex, fbs, exang, target
- **Categorical Features (encoded numerically):** cp, restecg, slope, ca, thal

All categorical variables are already encoded as integers, which makes the dataset easy to preprocess for machine learning models.

5. Data Quality

- The dataset does not contain missing values.
- All features are in numerical format.
- The dataset size is relatively small, which allows fast training and experimentation.

6. Suitability for Machine Learning Tasks

- **Classification:** Predicting the presence or absence of heart disease using the `target` column.
- **Regression:** Predicting a continuous variable such as `age`.
- **Clustering:** Identifying patient groups based on similar medical characteristics without using the target label.

7. Summary

The Heart Disease dataset is clean, well-structured, and highly suitable for educational machine learning projects. Its balanced structure and clear target variable make it an ideal choice for implementing and evaluating multiple machine learning models within a single project pipeline.